

Metals and Non-metals (Set A) — MCQ Worksheet

Science • Chapter 3 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Which metal is liquid at room temperature?

- (A) Sodium
- (C) Aluminium

- (B) Mercury
- (D) Iron

Q2. The most ductile metal is:

- (A) Copper
- (C) Gold

- (B) Aluminium
- (D) Iron

Q3. The best conductor of electricity among the following is:

- (A) Copper
- (C) Aluminium

- (B) Silver
- (D) Gold

Q4. Which non-metal is lustrous?

- (A) Sulphur
- (C) Phosphorus

- (B) Iodine
- (D) Carbon (charcoal)

Q5. The hardest natural substance is:

- (A) Iron
- (C) Diamond

- (B) Graphite
- (D) Quartz

Q6. Which of these is an amphoteric oxide?

- (A) CaO
- (C) Al_2O_3

- (B) Na_2O
- (D) CuO

Q7. Sodium is stored under:

- (A) Water
- (C) Kerosene

- (B) Alcohol
- (D) Oxygen

Q8. Which metal does NOT react with dilute HCl?

- (A) Zinc
- (C) Iron

- (B) Magnesium
- (D) Copper

Q9. Reaction of sodium with cold water is:

- (A) Slow
- (C) Endothermic

- (B) Vigorous and exothermic
- (D) Does not occur

Q10. In the activity series of metals, the most reactive is:

- (A) Copper
- (C) Potassium

- (B) Iron
- (D) Zinc

Q11. When a metal reacts with a non-metal, the bond formed is mostly:

- (A) Covalent
- (C) Metallic

- (B) Ionic
- (D) Hydrogen

Q12. Ionic compounds typically have:

- (A) Low melting points
- (C) No fixed structure

- (B) High melting points
- (D) Are gases at room temperature

Q13. Ionic compounds conduct electricity:

- (A) In solid state only
- (C) In molten or aqueous form

- (B) Always
- (D) Never

Q14. Roasting is heating a sulphide ore in:

- (A) Absence of air
- (C) Presence of carbon

- (B) Presence of excess air
- (D) Presence of water

Q15. Calcination is heating a carbonate ore in:

- (A) Excess air
(C) Water

- (B) Absence of air
(D) Hydrogen

Q16. Reduction of zinc oxide gives zinc when heated with:

- (A) Carbon
(C) Water

- (B) Oxygen
(D) Sulphur

Q17. Aluminium is extracted from its ore by:

- (A) Reduction with carbon
(C) Heating with hydrogen

- (B) Electrolysis of molten Al_2O_3
(D) Displacement using zinc

Q18. In electrorefining of copper, pure copper is deposited at:

- (A) Anode
(C) Electrolyte

- (B) Cathode
(D) Container wall

Q19. Rusting of iron requires:

- (A) Only oxygen
(C) Both oxygen and water

- (B) Only water
(D) Neither oxygen nor water

Q20. Galvanisation involves coating iron with a layer of:

- (A) Tin
(C) Zinc

- (B) Aluminium
(D) Chromium

Q21. Stainless steel is an alloy of iron with:

- (A) Cu, Zn
(C) Pb, Sn

- (B) Cr, Ni, C
(D) Au, Ag

Q22. Brass is an alloy of:

- (A) Cu and Sn
(C) Pb and Sn

- (B) Cu and Zn
(D) Fe and Cr

Q23. Bronze is composed of:

- (A) Cu + Zn
(C) Cu + Ni

- (B) Cu + Sn
(D) Fe + C

Q24. 24-carat gold means:

- (A) 100% pure gold
(C) 75% pure gold

- (B) 91.6% pure gold
(D) 50% pure gold

Q25. The non-metal that conducts electricity is:

- (A) Sulphur
(C) Graphite

- (B) Diamond
(D) Iodine